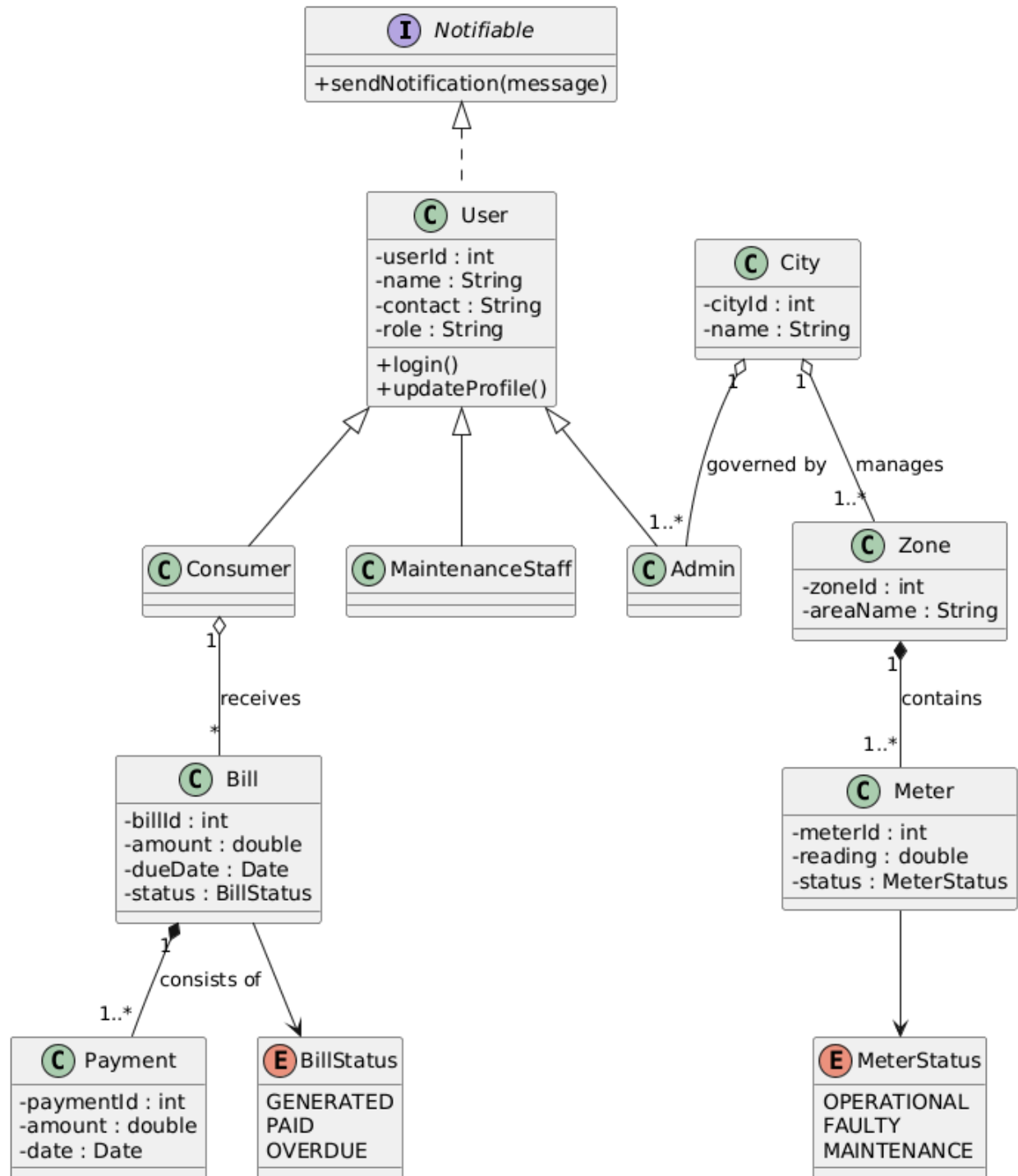




Classes Identified

User (Abstract Class)

Attributes:

- userId : int
- name : String
- email : String
- phone : String
- password : String

Methods:

- login()
- logout()
- updateProfile()

Consumer

Attributes:

- consumerId : int
- address : String

Methods:

- viewBills()
- makePayment()
- viewConsumption()

MaintenanceStaff

Attributes:

- staffId : int
- department : String

Methods:

- assignTask()
- updateTaskStatus()

CityOfficial

Attributes:

- officialId : int
- designation : String

Methods:

- generateReport()
- updateTariffPolicy()

DistributionZone

Attributes:

- zoneId : int
- zoneName : String
- location : String

Methods:

- addMeter()
- monitorZone()

WaterMeter

Attributes:

- meterId : int
- installationDate : Date
- status : MeterStatus

Methods:

- recordReading()
- updateStatus()

MeterReading

Attributes:

- readingId : int
- readingDate : Date
- consumption : double

Methods:

- saveReading()

Bill

Attributes:

- billId : int
- billDate : Date
- amount : double
- dueDate : Date
- status : BillStatus

Methods:

- generateBill()
- calculateAmount()

Payment

Attributes:

- paymentId : int
- paymentDate : Date
- amount : double
- paymentMethod : String

Methods:

- processPayment()

TariffPolicy

Attributes:

- policyId : int
- slabRate : double

Methods:

- updateRate()

MaintenanceTask

Attributes:

- taskId : int
- description : String
- taskStatus : TaskStatus

Methods:

- assignTask()
- closeTask()

Alert

Attributes:

- alertId : int
- message : String
- alertDate : Date

Methods:

- sendAlert()

Report

Attributes:

- reportId : int
- reportType : String
- generatedDate : Date

Methods:

- generate()
-

5. Interface Identified

Notifiable

Method:

- sendNotification()

Implemented By:

- Alert
-

6. Enumerations Identified

MeterStatus

- OPERATIONAL
- FAULTY
- UNDER_MAINTENANCE

BillStatus

- PAID
- UNPAID
- OVERDUE

TaskStatus

- ASSIGNED
 - IN_PROGRESS
 - COMPLETED
-

7. Inheritance (Generalization)

User

- Consumer
 - MaintenanceStaff
 - CityOfficial
-

8. Relationships with Multiplicity

Aggregation

- DistributionZone (1) ◇—— (1..*) WaterMeter

Composition

- WaterMeter (1) ◆—— (1..*) MeterReading
- Bill (1) ◆—— (0..1) Payment

Associations

- Consumer (1) —— (1..*) WaterMeter
- Consumer (1) —— (0..*) Bill
- Consumer (1) —— (0..*) Payment
- MaintenanceStaff (1) —— (0..*) MaintenanceTask
- MaintenanceTask (*) —— (1) WaterMeter
- CityOfficial (1) —— (0..*) Report
- CityOfficial (1) —— (1) TariffPolicy
- Consumer (1) —— (0..*) Alert
- MaintenanceStaff (1) —— (0..*) Alert
- CityOfficial (1) —— (0..*) Alert